

Why we use Code First rather than Database First?

Code First lets us define the database using C# classes, so the code is the single source of truth.

It is flexible, allows easy changes through migrations, and is ideal for new projects.

Database First is only used when an existing database already exists.

Why we use Fluent API over Data Annotations?

Fluent API provides more control and flexibility for configuring entities and relationships.

It keeps classes clean by centralizing configuration in the DbContext.

Data Annotations are simpler but limited to basic rules like [Required] or [MaxLength].

Self Study:

Class Diagram Specs

Class diagram shows **all entities, their properties, and relationships** in the system.

It helps visualize the structure of the database before and after mapping classes using EF Core.

Up to New Change vs Up to Previous Change

- **Up to New Change** → Includes **all migrations and updates** from the beginning to the latest change.
- **Up to Previous Change** → Includes **only migrations up to the previous step**, ignoring the latest migration.

Useful when you want to **rollback or compare database versions**.

One Line for All Configurations

Fluent API allows you to configure **all entity properties, relationships, and constraints in one place** (OnModelCreating) instead of using scattered attributes.